Brain-heart infusion (BHI) agar with antibiotics and actidione at 25°C and 37°C. On Sabourad dextrose agar the colonies appear albino or brown. The albino type consists of white, fine aerial hyphae and brown type consists of flat colonies with light tan or dark brown in color in seven days. At 37°C the colonies grow as granular to rough, mucoid and cream-colored turning tan to brown in 14 days.

ii. Treatment

Amphotericin B is given for the treatment of disseminated and other severe forms of Histoplasmosis.

9.5 Opportunistic Mycoses

The opportunistic systemic mycoses are infections found in patients with underlying pre disposing conditions. It is produced by non pathogenic or contaminant fungi in a host, where the immunological defense mechanisms are weakened by endogenous causes like cancer, leukemia or exogenous causes like immunosuppressive therapy and AIDS. The examples of opportunistic mycoses are Candidiasis, Cryptococcosis, Aspergillosis and zygomycosis.

9.5.1 Candidiasis

Candidiasis is the commonest fungal disease found in humans affecting **mucosa, skin, nails and internal organs** of the body. It is caused by yeast like fungi called *Candida albicans*. The infection may be acute or chronic, superficial or deep and found mainly as secondary infection in individuals with immune compromised condition.



The **fungus *candida albicans*** is responsible for most **vaginal** yeast infections. Your vagina naturally contains a balanced mix of yeast, including candida, and bacteria. Certain bacteria (lactobacillus) act to prevent an **overgrowth** of yeast. But that balance can be disrupted.

Pathogenesis and Pathology

Some of the virulence factors contributing to pathogenicity are toxins, enzymes and adhesion. The organism adheres to the epithelial and endothelial cells by proteinase production. Then the yeast cells of *Candida* encounter a particular host tissue and colonization takes place at the local site or they invade deeper into the host tissue and induce various clinical symptoms.

Clinical Features

The *Candida* species are found as commensal on mucosal surfaces of the body. They cause disease as and when conditions are favourable. This yeast like fungi colonizes mucocutaneous surfaces, which can be portals of entry into deeper tissues when the host defenses are compromised. They may cause a simple lesion to event the life threatening systemic infection.

The clinical manifestations of Candidiasis are divided into two broad categories. They are:

1. Infectious Diseases

a. Mucocutaneous Involvement

i. Oral Candidiasis – Most common form *Candida* colonizes on the oral

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